Representation Varieties and Fixed Points

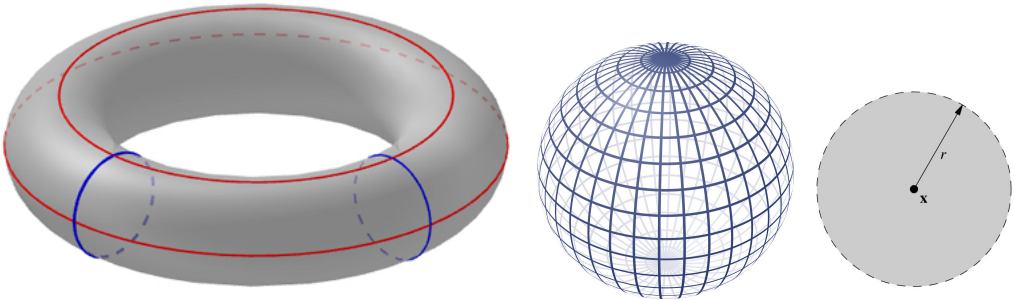
Timothy Tarter
Joint With Patrick Dendis, David Duncan, Lior Weber, Colin Willette

James Madison University

November 7, 2026

What Are We Trying To Do?

- A surface is a space that looks locally 2-dimensional.



- We're interested in results that arise from transformations of surfaces.
- Some of the information from these transformations is encoded in an interesting space $\mathcal{R}$, which is interesting to particle physicists.

The Group S^3

- The quaternions, $\mathbb{H}$, are an extension of the complex numbers.





- We consider S^3 as the set of all quaternions with length 1.

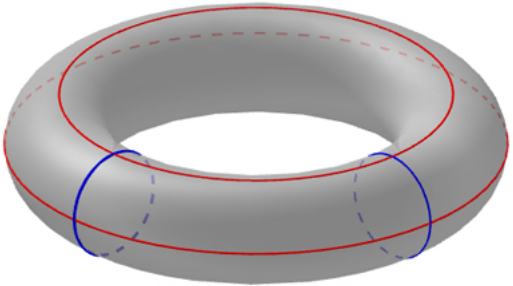
$$S^3 = \{a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k} \in \mathbb{H} \mid a^2 + b^2 + c^2 + d^2 = 1\}$$

Representation Varieties

- For the torus T^2, the **representation variety** is the set

$$\mathcal{R}(T^2) = \{(A, B) \in (S^3)^2 \mid AB = BA\}$$

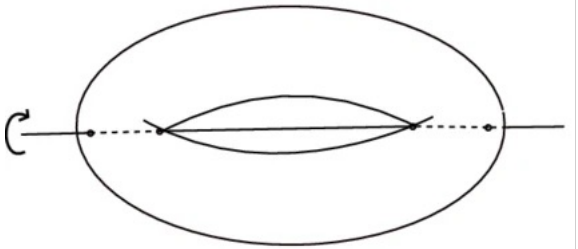
- For a general surface Σ, the representation variety $\mathcal{R}(\Sigma)$ is a space, so we can consider properties of $\mathcal{R}(\Sigma)$ such as connectedness.



Transformations of a Surface

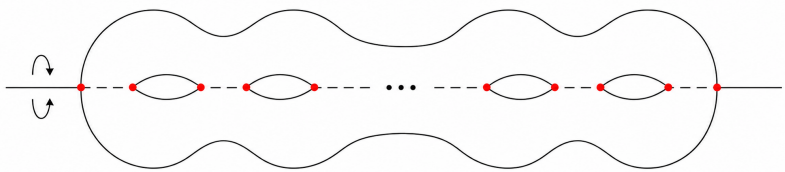
Flipping over a torus induces an action on the representation variety. The fixed points under this action are

$$\{(A, B) \in (S^3)^2 \mid A, B \in \{\pm 1\}\} = \{\pm 1\}^2$$



Flipping a Surface Over - Transformations of Order 2

For a surface with g holes, we can think about a similar transformation.



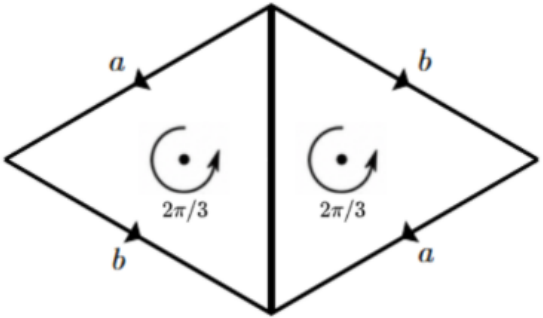
Theorem 1

The set of fixed points of this action on $\mathcal{R}(\Sigma_g)$ is homeomorphic to

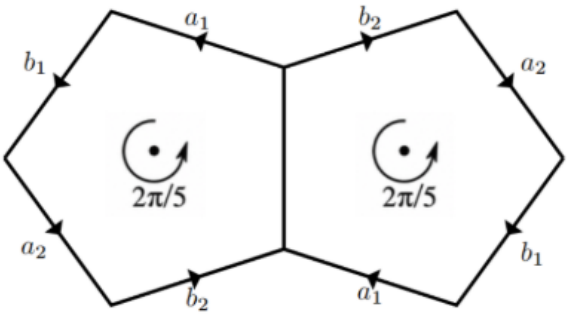
$$\{(A_1, A_2, ..., A_{2g}) \in (S^3)^{2g} \mid A_1, A_2, ..., A_{2g} \in \{\pm 1\}\} = \{\pm 1\}^{2g}$$

Transformations of Order $2g + 1$

One particular transformation of the surface with g holes that we are interested in is one of order $2g + 1$.



One hole



Two holes

Transformations of Order $2g+1$

Theorem 2

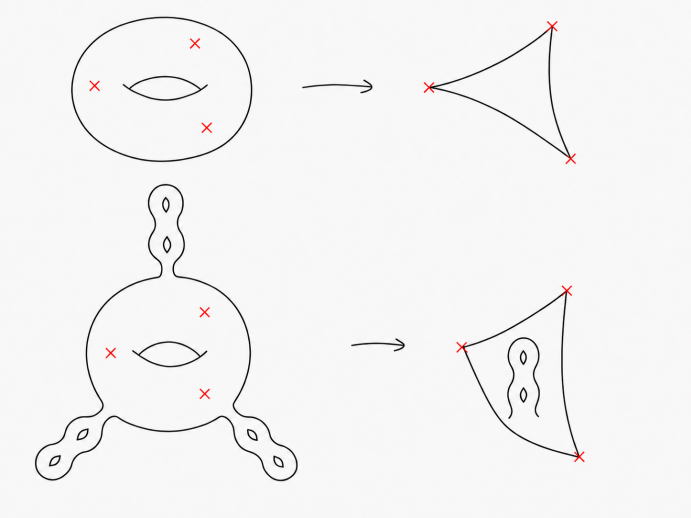
The set of fixed points of this action on $\mathcal{R}(\Sigma_g)$ is homeomorphic to

$$\{1\} \sqcup \left(\bigsqcup_{i=1}^{g} S^2 \right).$$

The set of fixed points of this action on $\mathcal{R}(N_{g,k})$ is homeomorphic to

$$\{1\} \sqcup \left(\bigsqcup_{i=1}^{kg+\lfloor (k-1)/2 \rfloor} S^2 \right).$$

Extending Transformations of Order 2g+1



Extending Transformations of Order 2g+1

Theorem 3

The fixed point set of $R(\Sigma_{g+(2g+1)\ell})$ under ϕ is

$$\{(A, C, D) \in (S^3)^3 \mid A^{2g+1} = 1, [C, D]^{\ell(2g+1)} = 1\} \tag{1}$$

and the number of connected components is

$$(g+1)\left(\left\lfloor \frac{\ell(2g+1)}{2} \right\rfloor + 1\right). \tag{2}$$

We have proven a similar result for all circle bundles over $\Sigma_{g+(2g+1)\ell}$.

Acknowledgments

We would like to give thanks to the NSF (grant DMS-2349593), the Elizabeth McConnell Bliss Endowment, and the Joan and Ernest Droms Endowment for their funding. We thank Dr. David Duncan for his gracious mentorship, along with the James Madison University Math and Stats Department. We also thank the University of Florida, Georgia Institute of Technology, and Carleton College.

ArXiV link will be available soon, see www.tartermathematics.com for updates.